Artificial Intelligence

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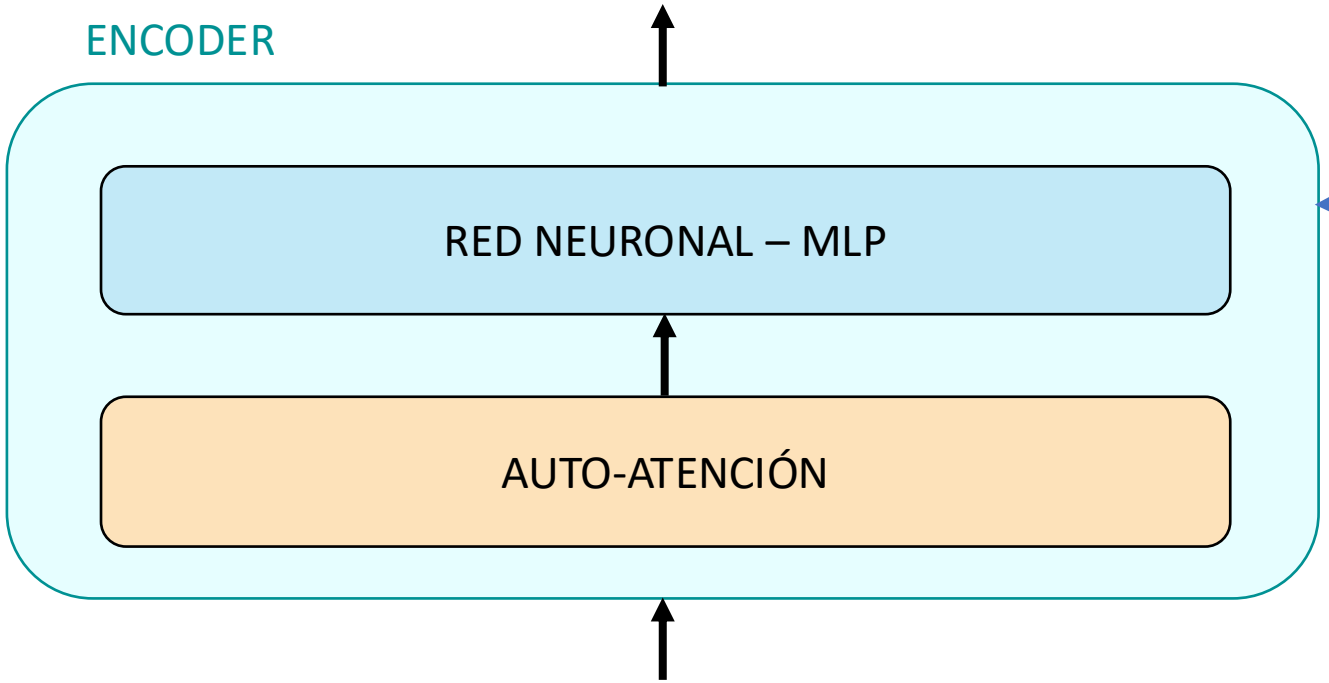
Facultad de Ingeniería en Electricidad y Computación – FIEC

Clase # 19 - PAO 2026 – I

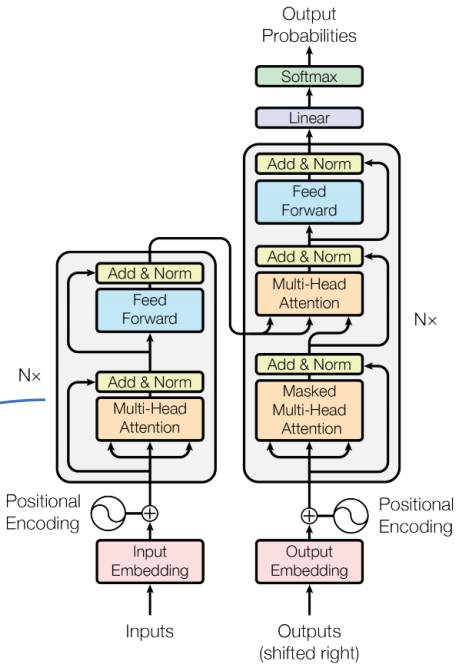
Agenda

1. Transformers:
    1.1 Auto-atenttion
    1.2 Las matrices: K, Q, V
    1.3 Multi-heads
    1.4 Embeddings: Ejemplos sobre embedddings: Palabras, posición

Transformers



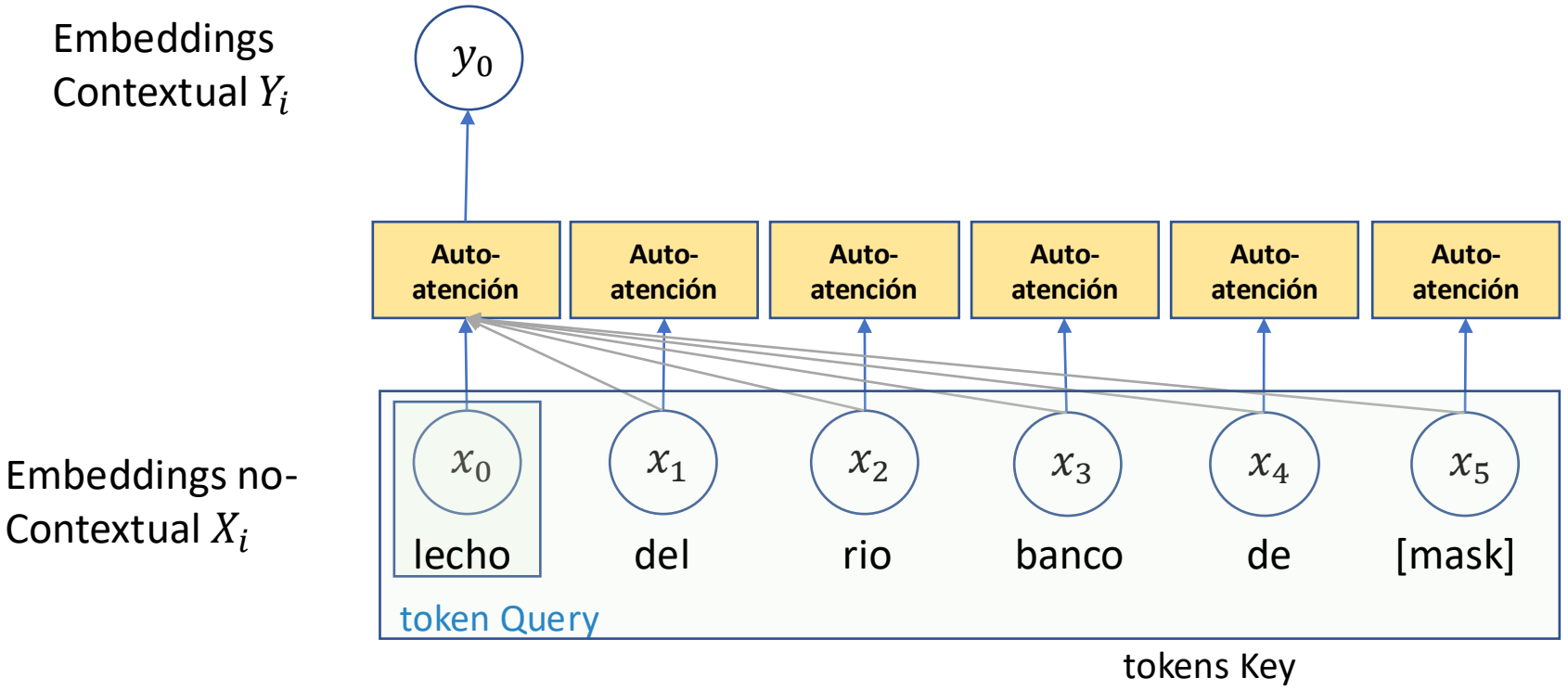
6 capas como esta



- Encoders son identicos, pero no comparten los pesos

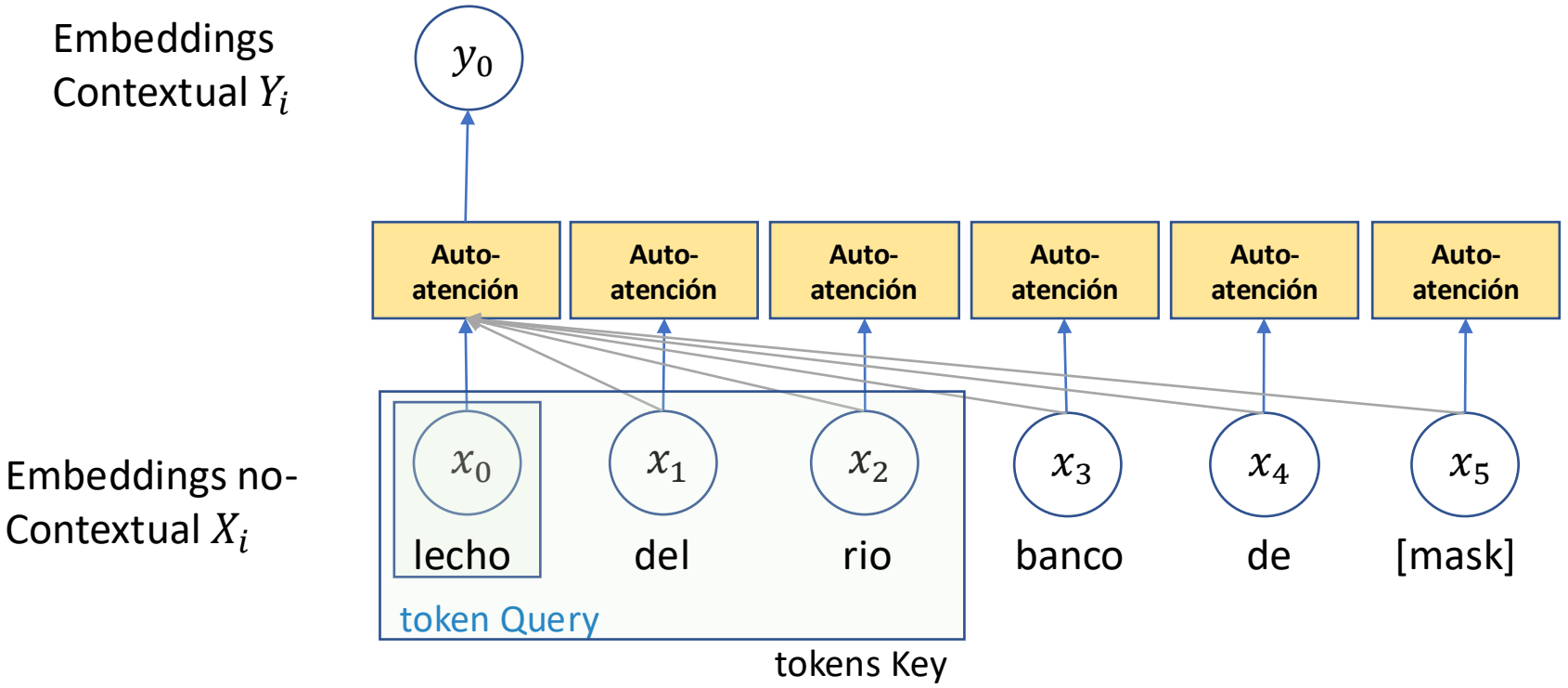
Transformers

- Auto-atención



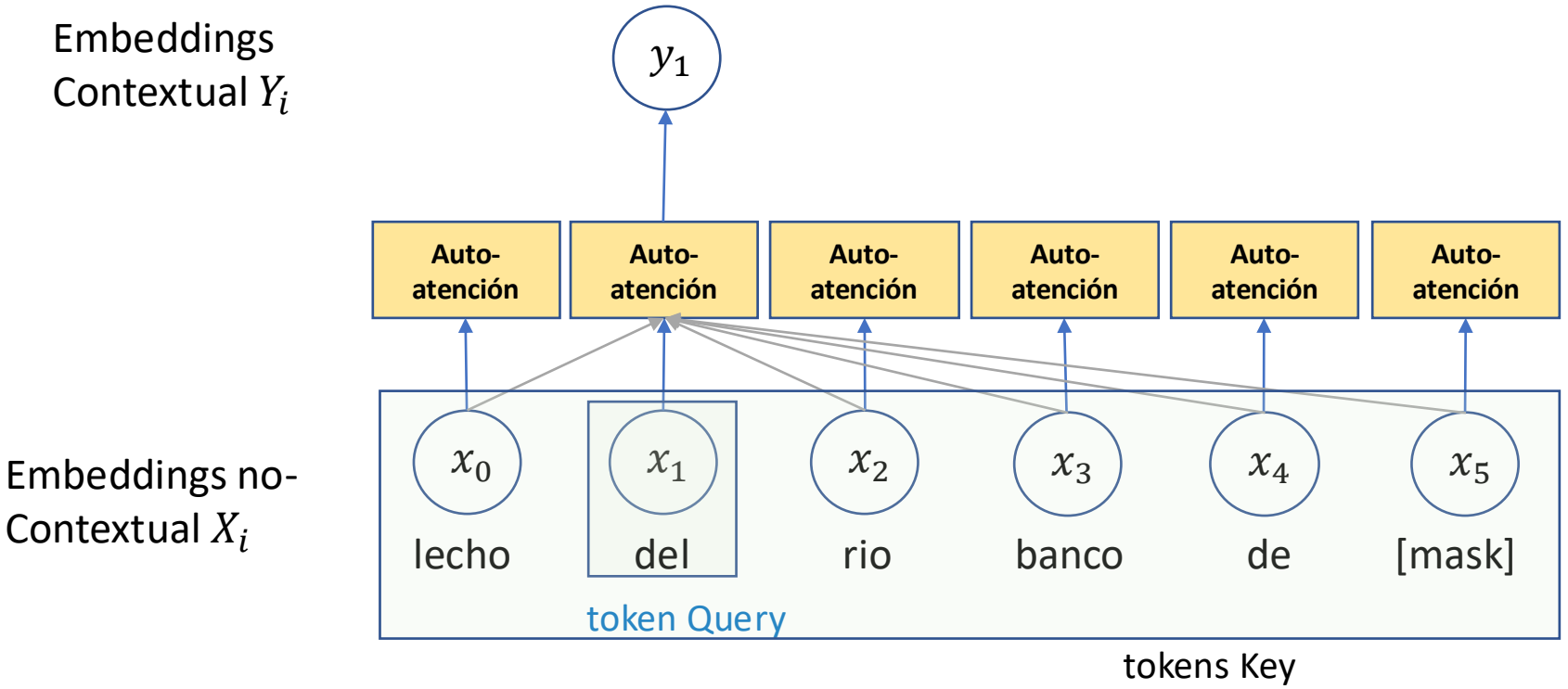
Transformers

- Auto-atención



Transformers

- Auto-atención

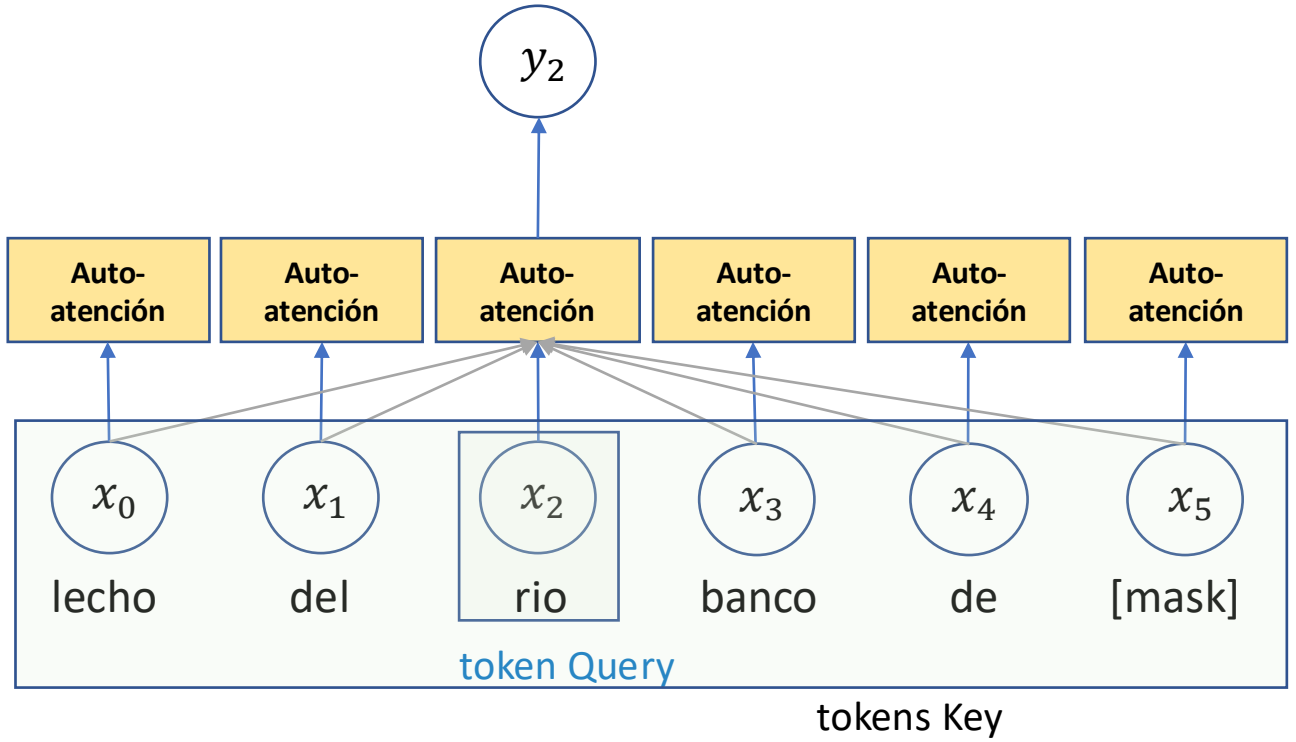


Transformers

- Auto-atención

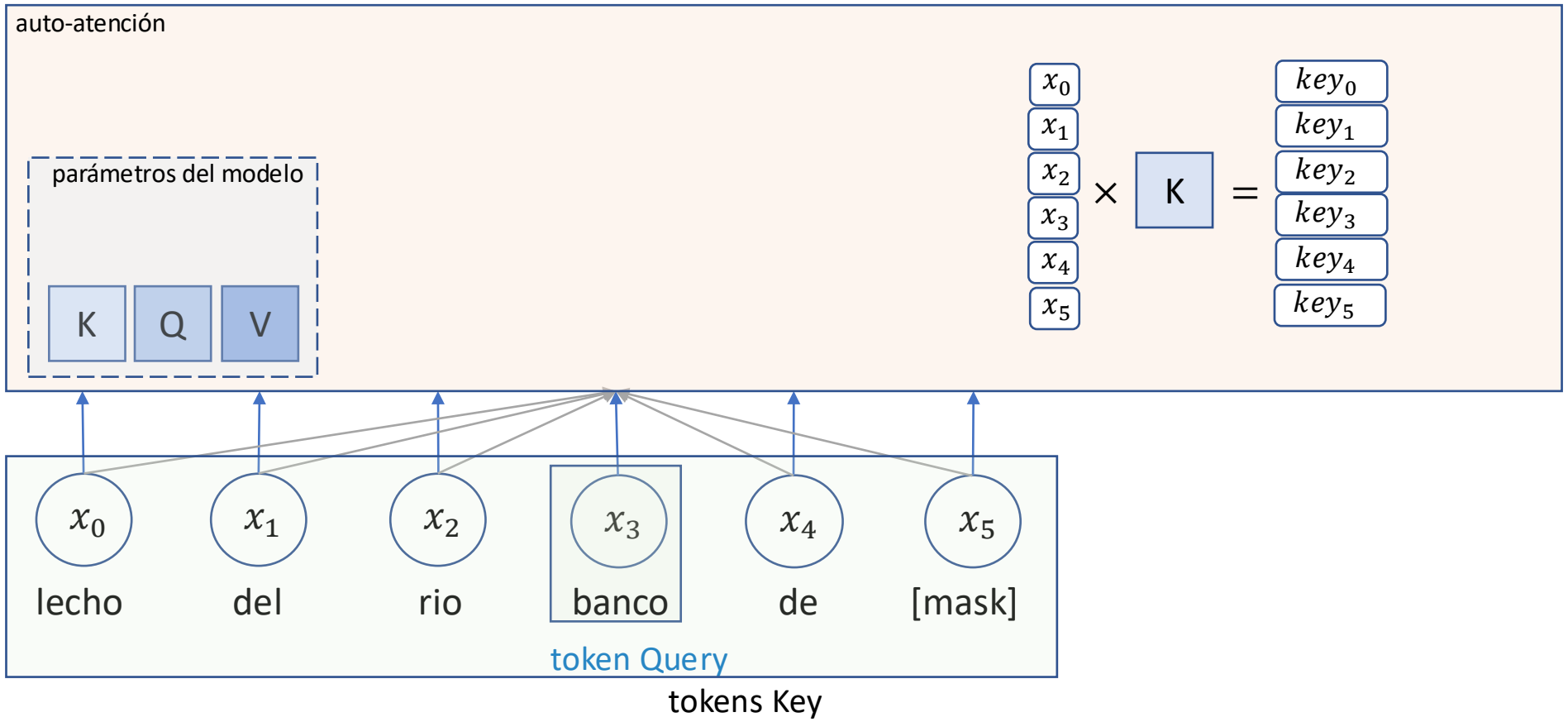
Embeddings Contextual Y_i

Embeddings no-Contextual X_i



Transformers

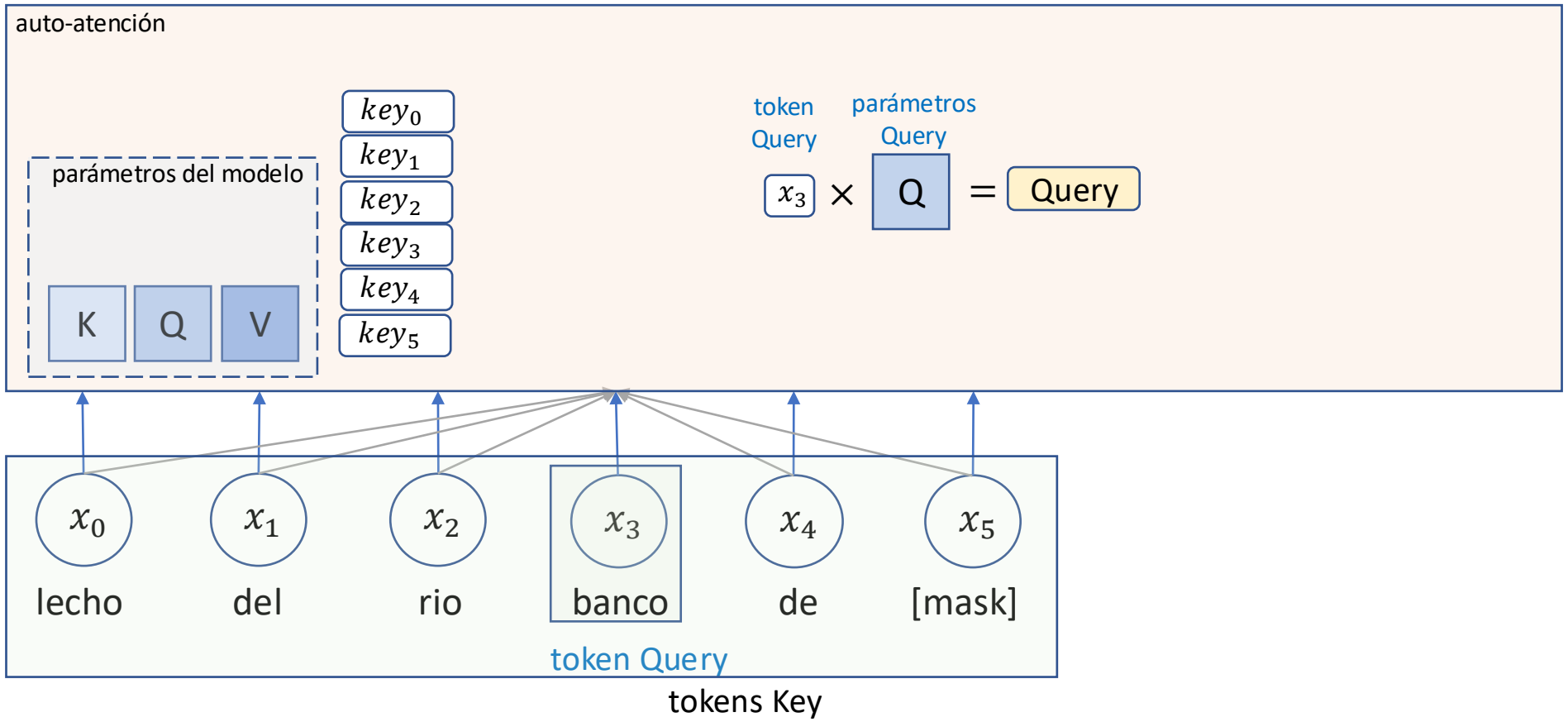
- Las Claves (Keys)



Embeddings no-Contextual X_i

Transformers

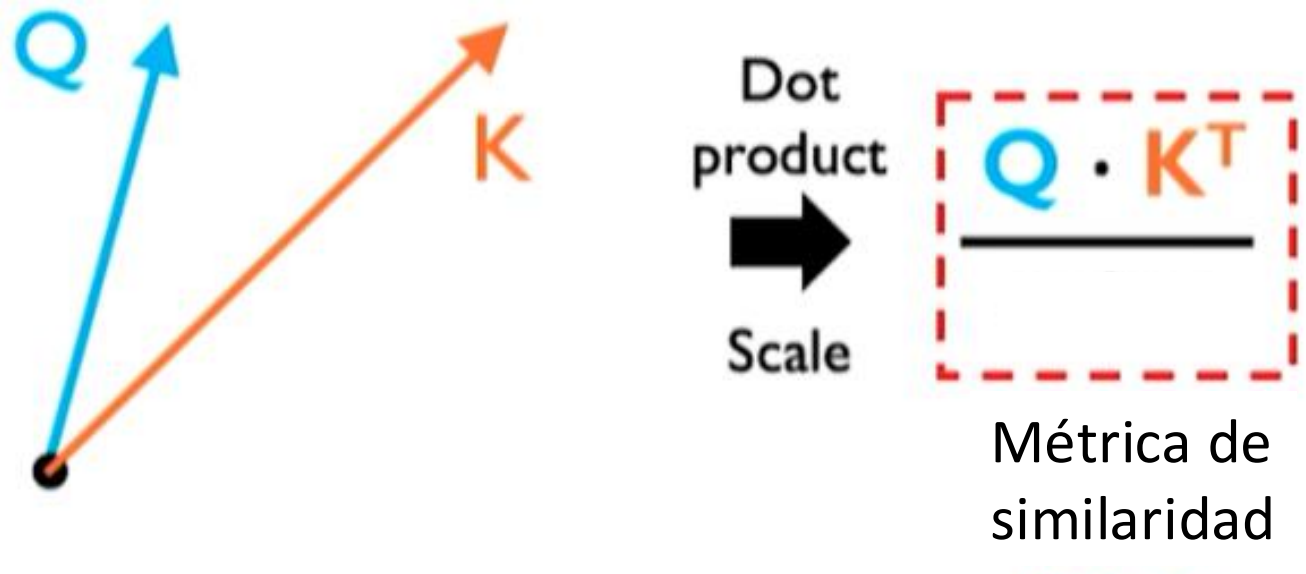
- Query



Embeddings no-Contextual X_i

Transformers

- Similitud de Q en K

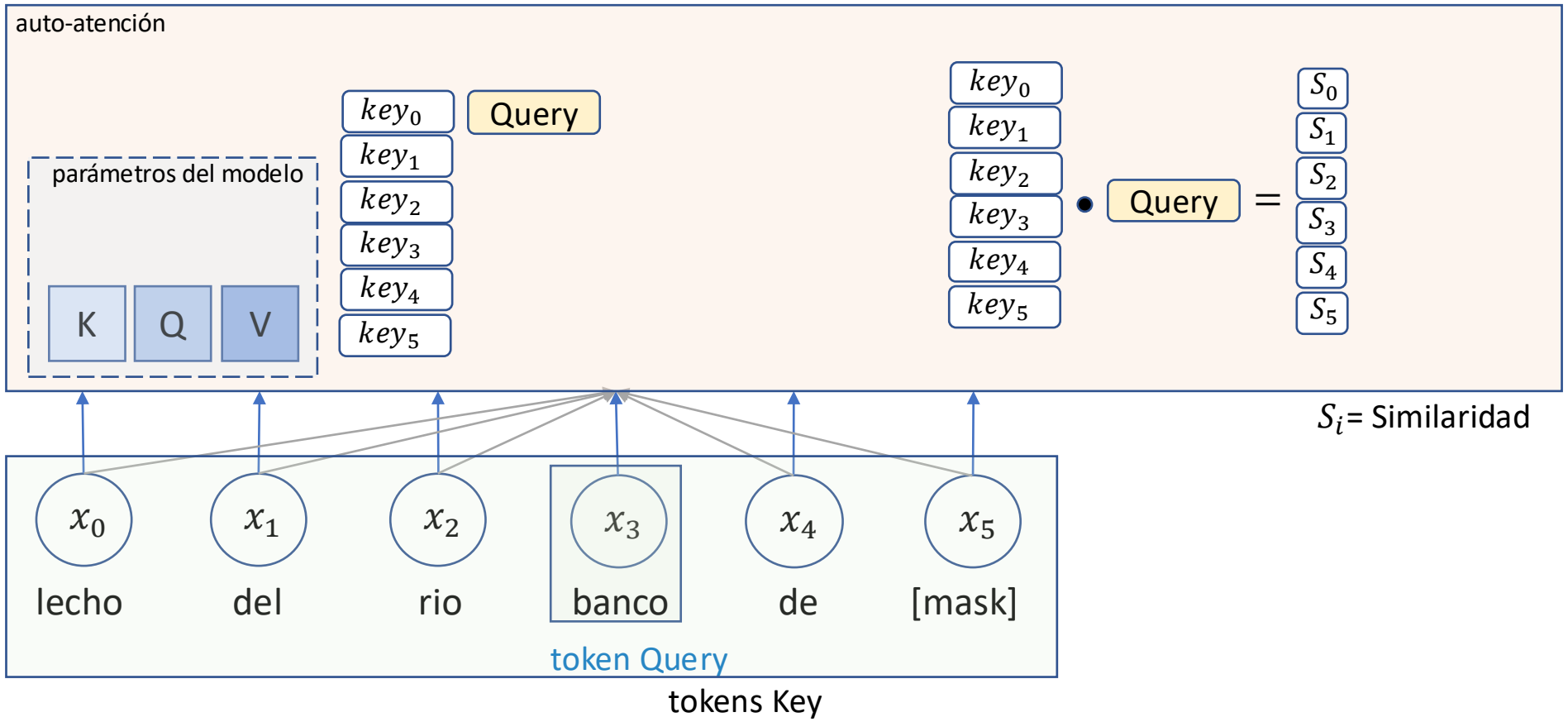


Métrica de similaridad

También conocido como el Coseno de Similaridad

Transformers

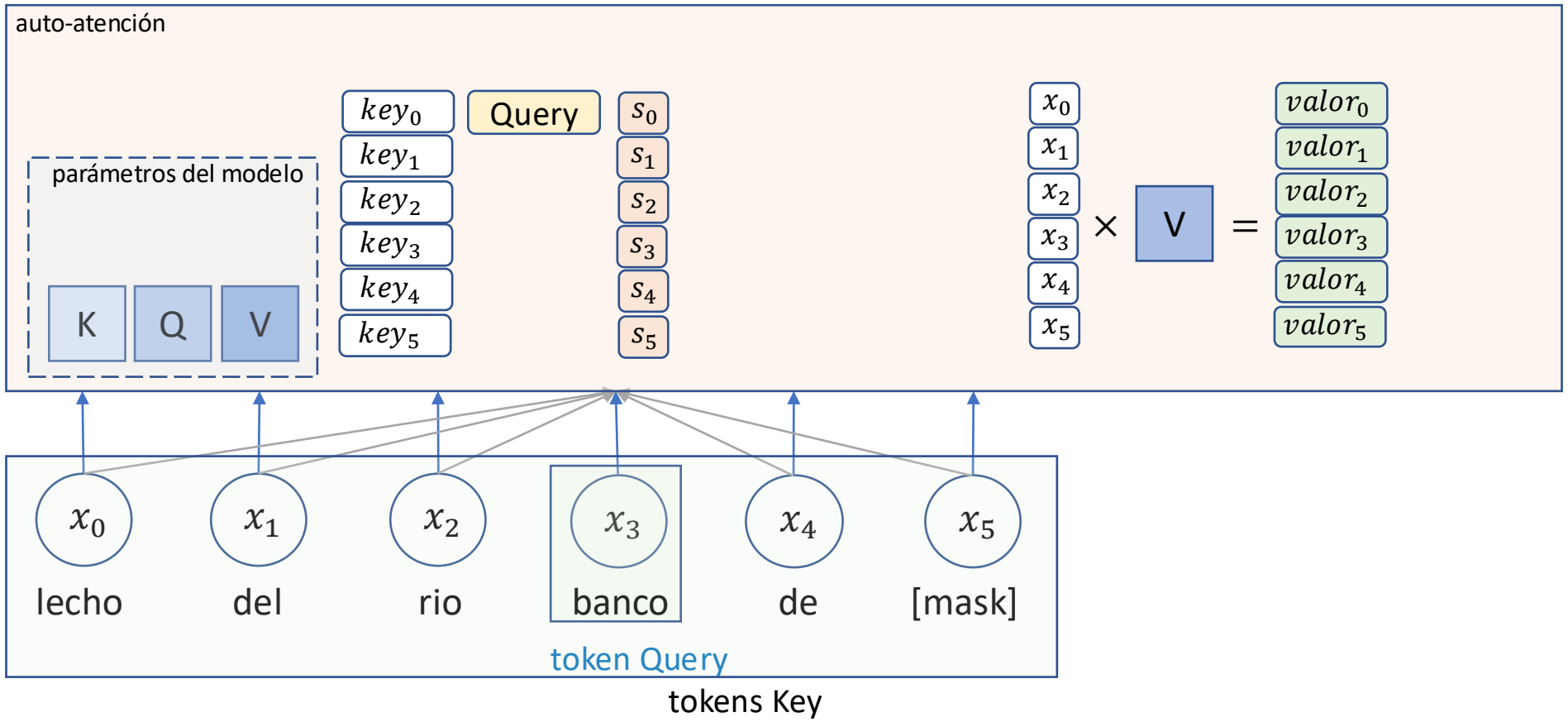
- Similitud



Embeddings no-Contextual X_i

Transformers

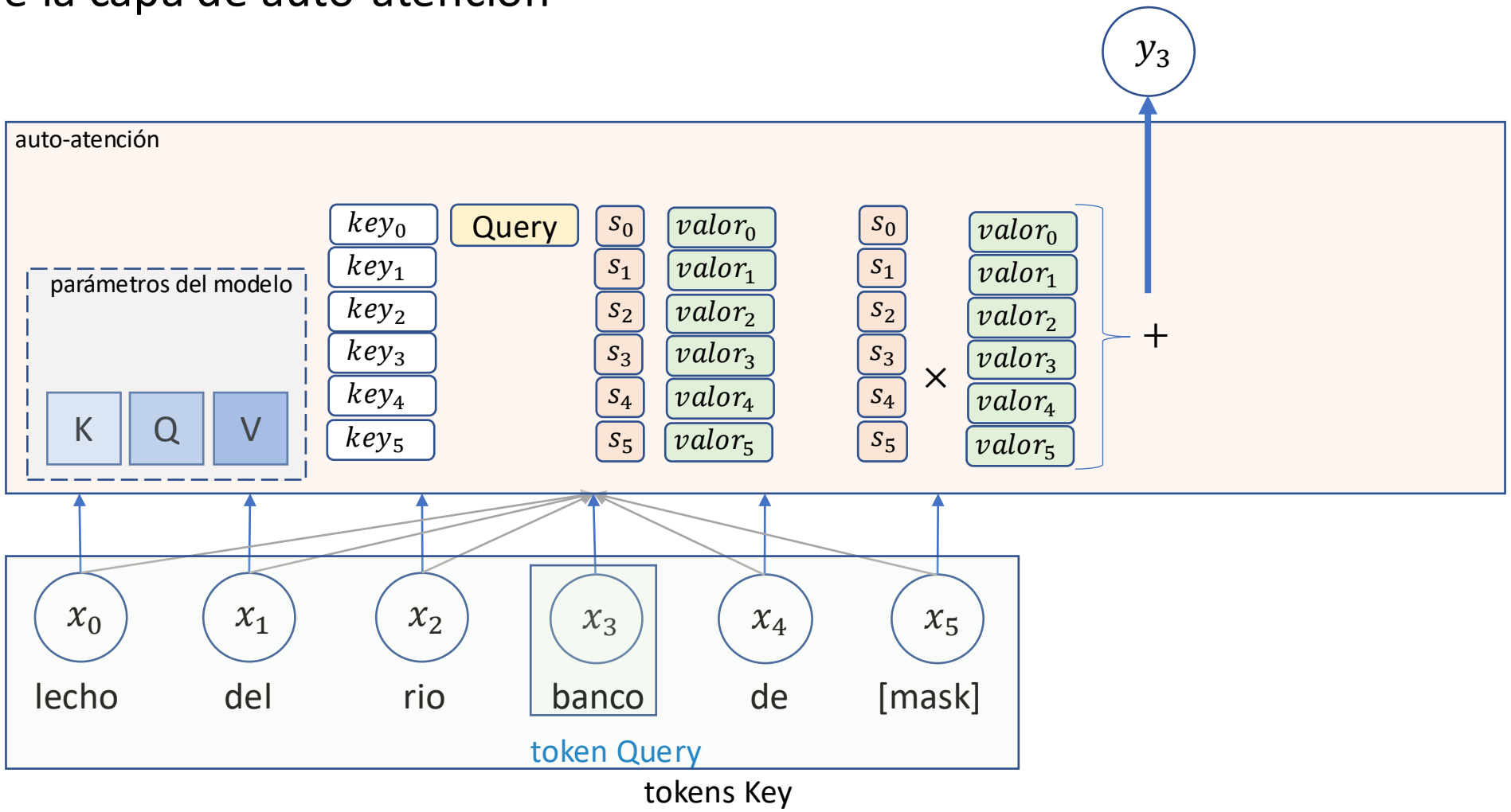
- Los Valores



Embeddings no-Contextual X_i

Transformers

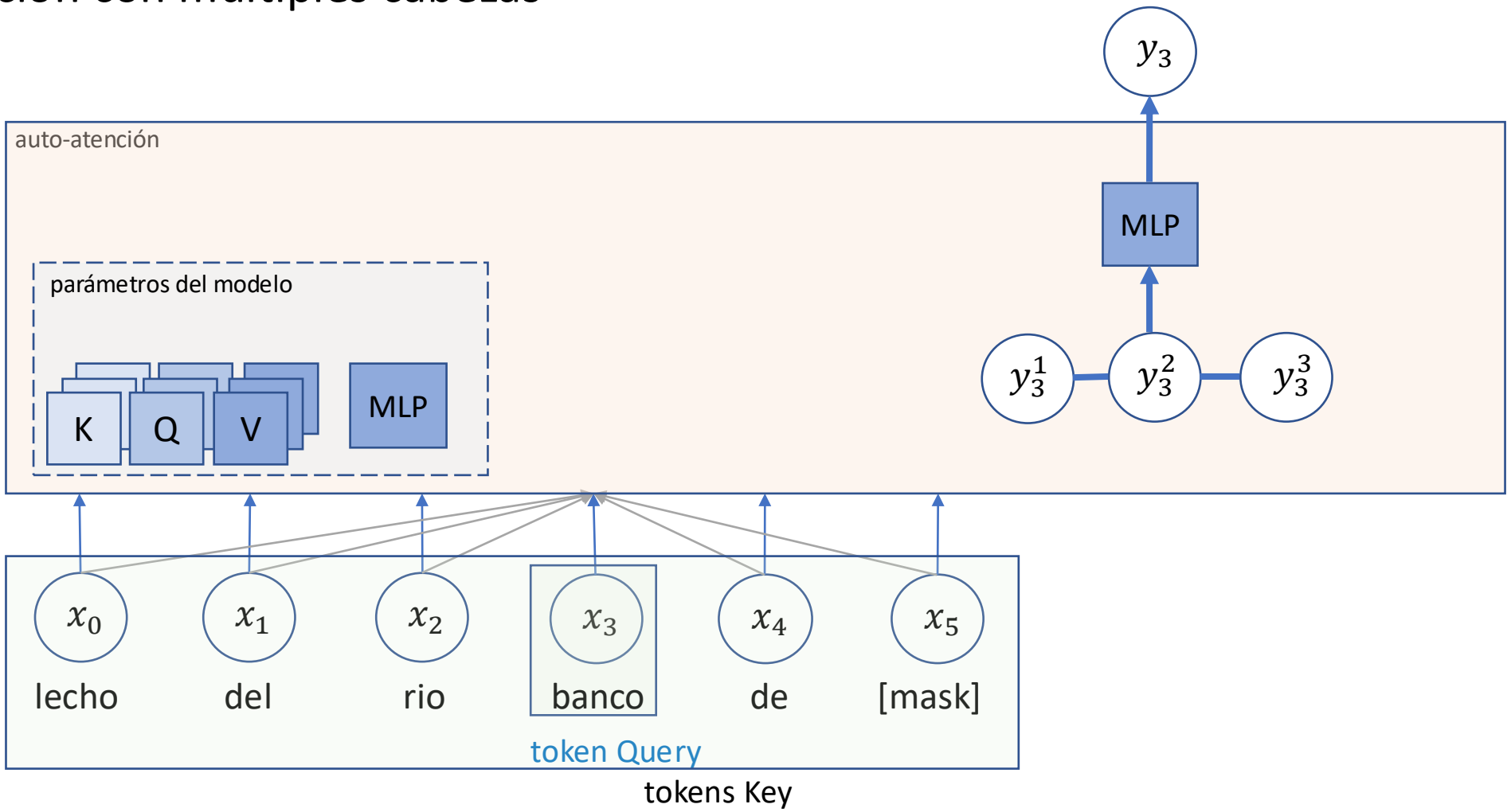
- La Salida de la capa de auto-atención



Embeddings no-Contextual X_i

Transformers

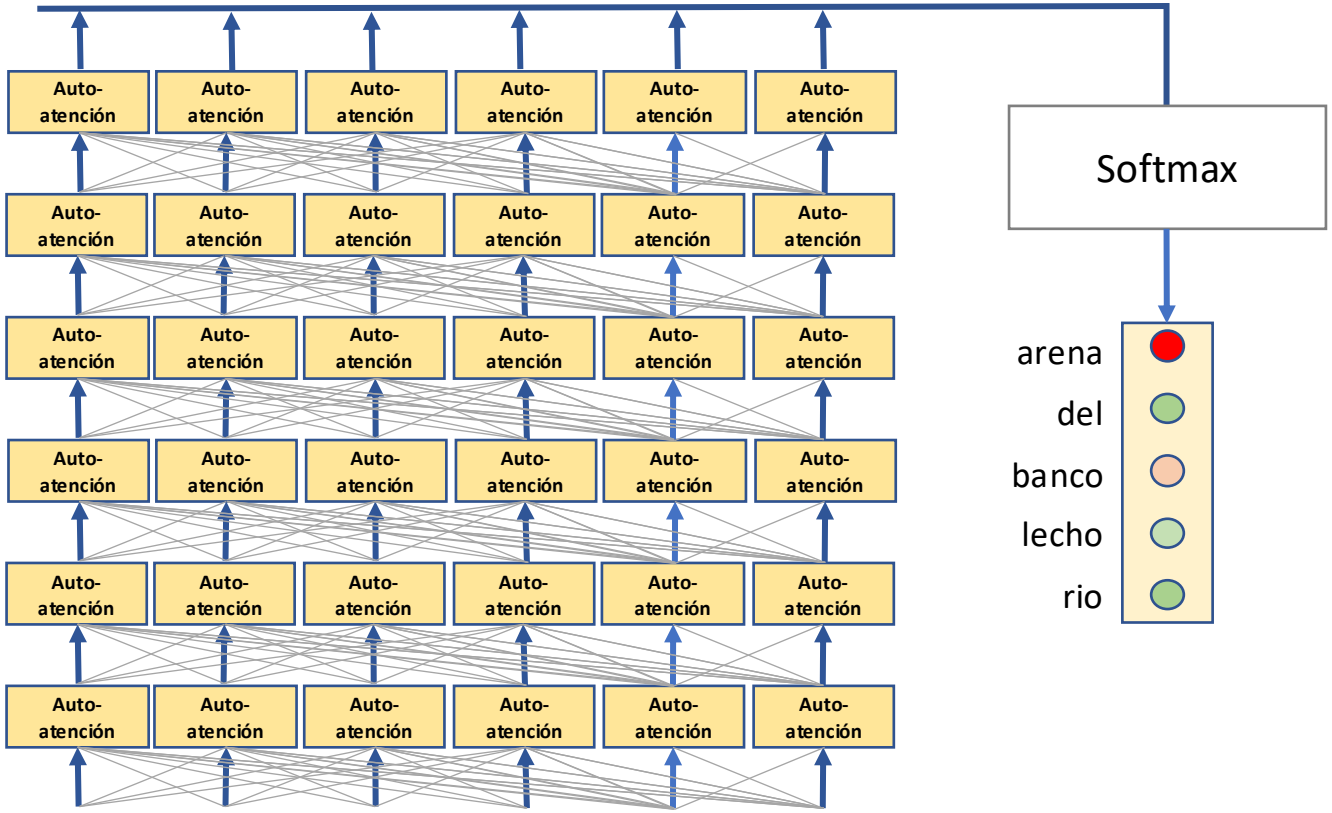
- Auto-atención con múltiples cabezas



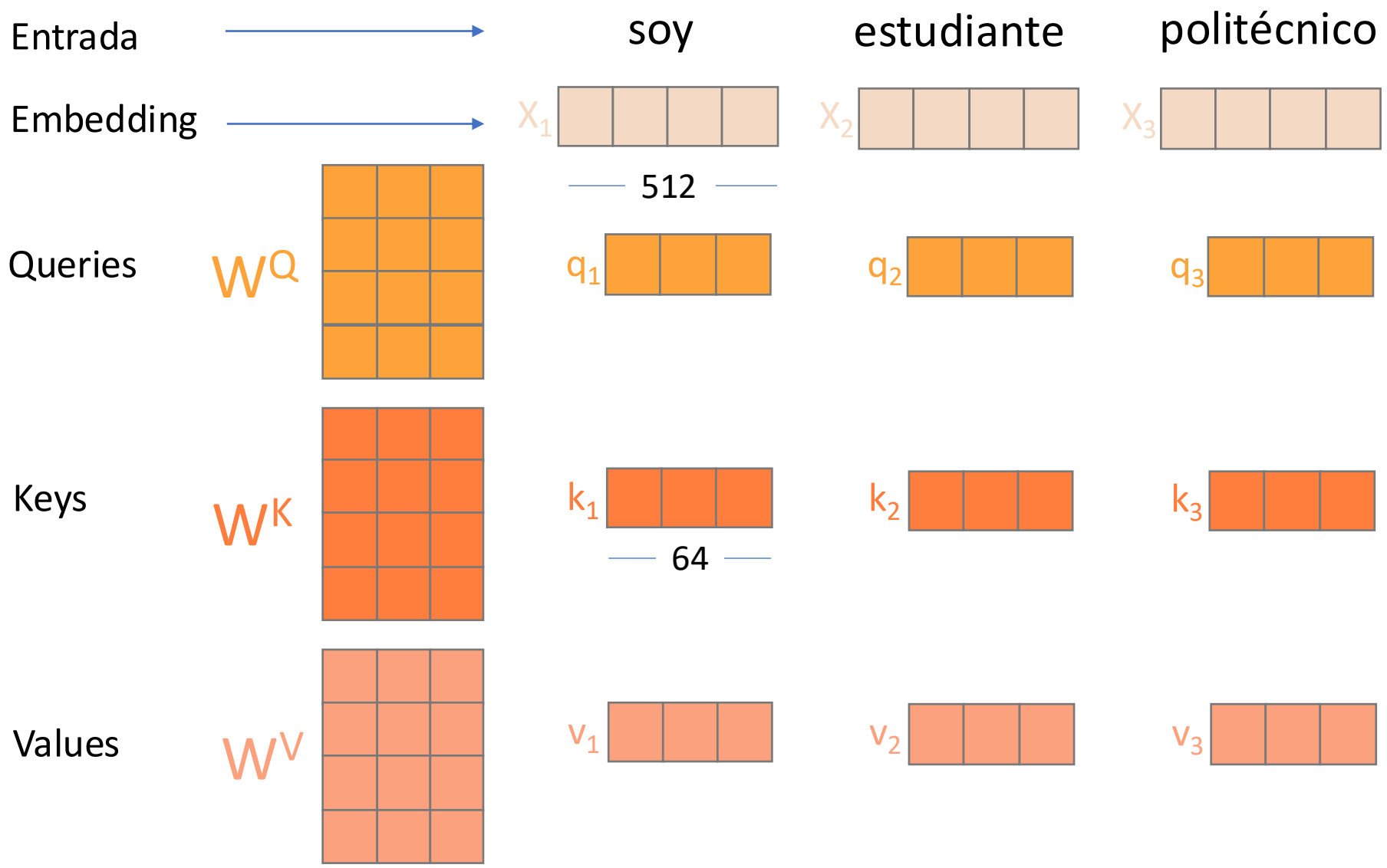
Embeddings no-Contextual X_i

Transformers

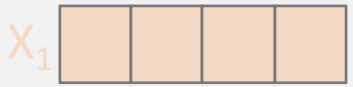
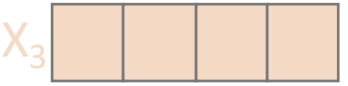
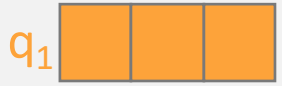
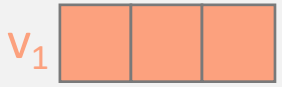
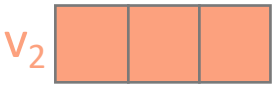
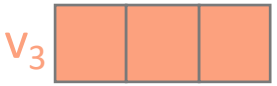
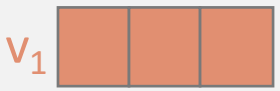
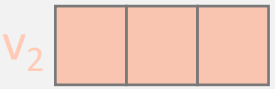
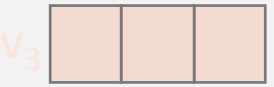
- El codificador - Transformer



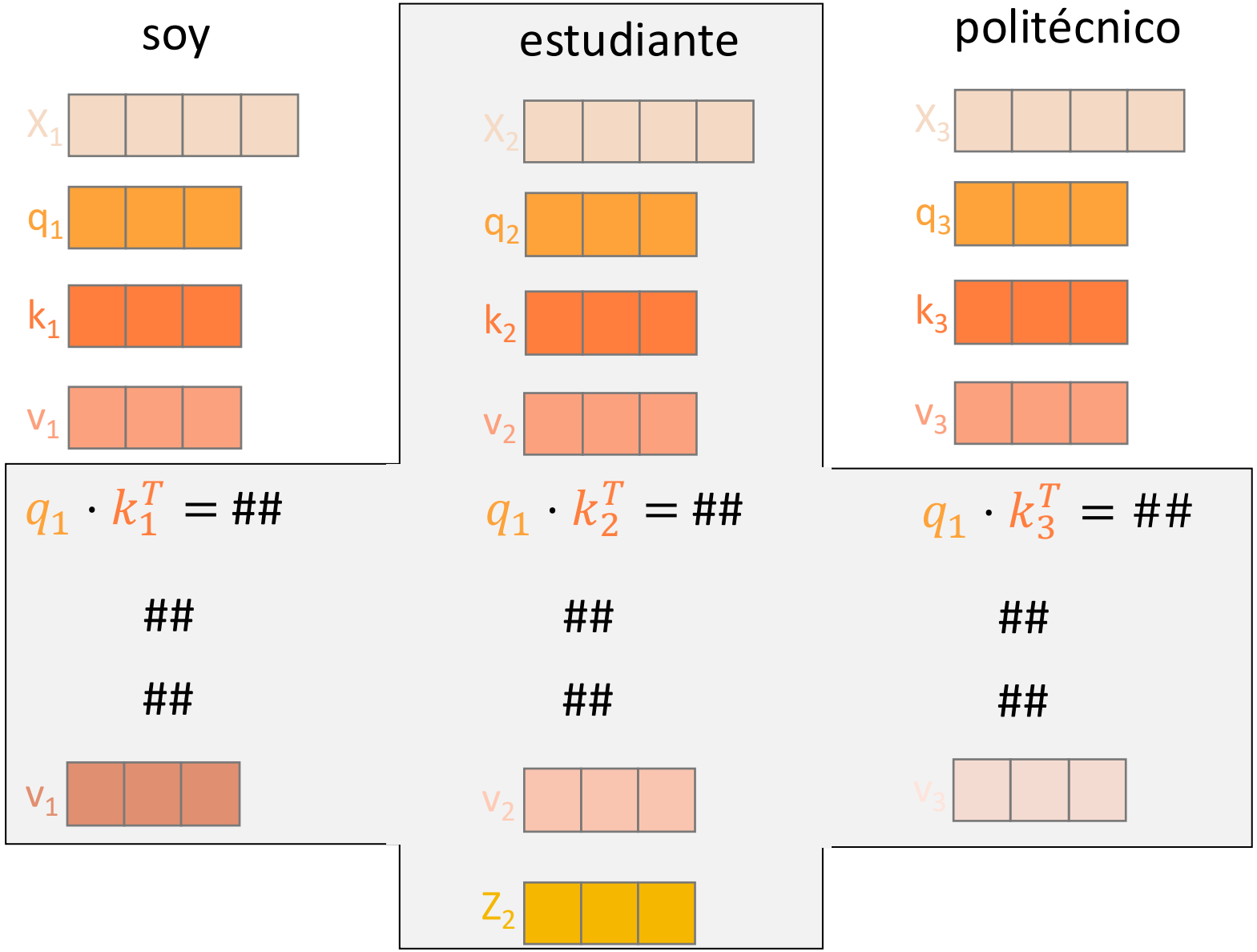
Transformers



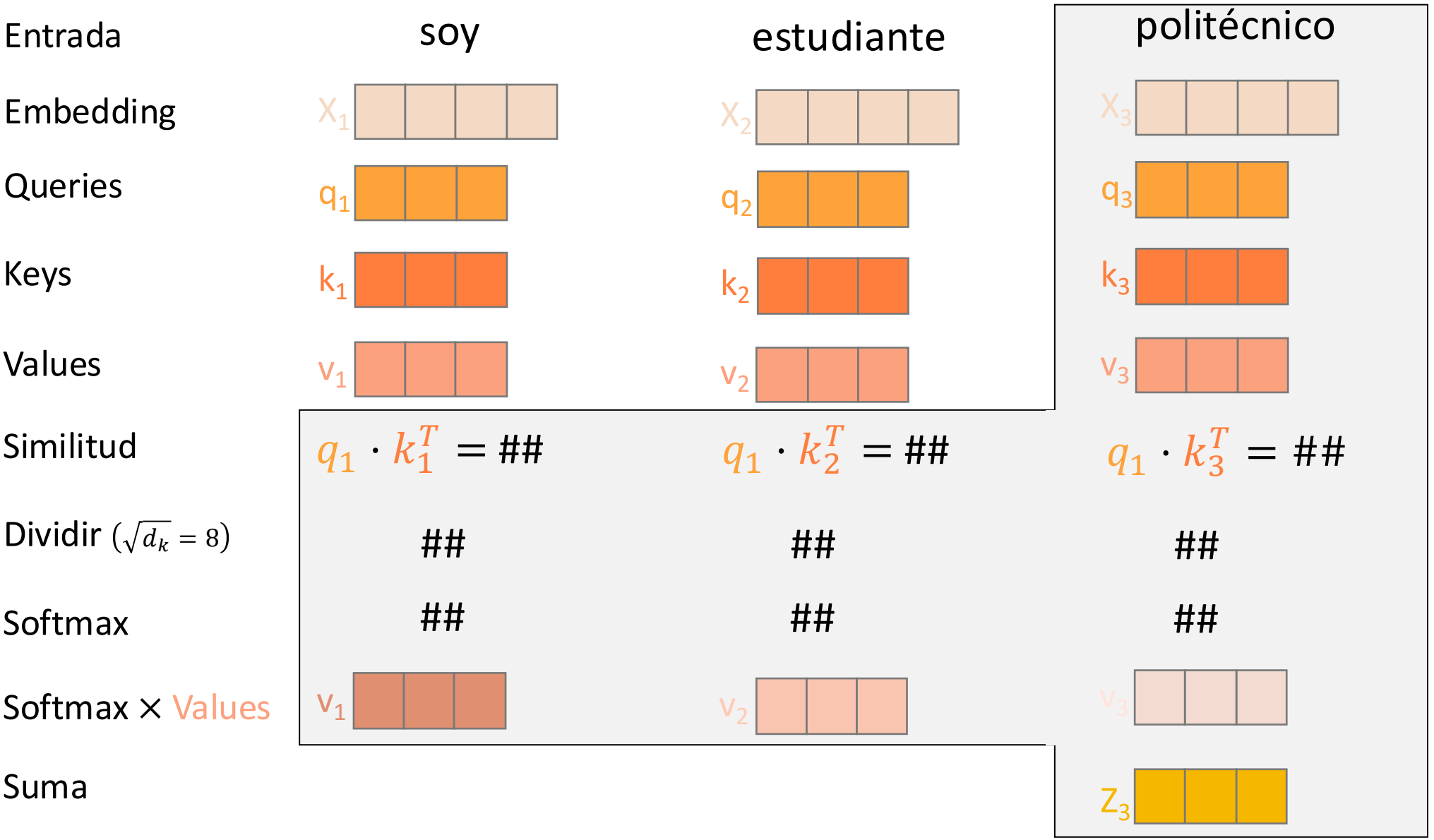
Transformers

Entrada	soy	estudiante	politécnico
Embedding			
Queries			
Keys			
Values			
Similitud	$q_1 \cdot k_1^T = 120$	$q_1 \cdot k_2^T = 82$	$q_1 \cdot k_3^T = 61$
Dividir $(\sqrt{d_k} = 8)$	15	10,3	7,6
Softmax	0,94	0,65	0,48
Softmax × Values			
Suma			

Transformers



Transformers



Transformers

